A PERFECT IMMERSION INTO A LOCALLY NOETHERIAN TARGET WITH FINITE LOCALLY FREE CONORMAL SHEAF IS REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18442-18451 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18453-18456, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Closes the circle opened in wave 12 with the perfect pair L16092/L16093 and wave 14's regular-immersion-perfect L16121: the reverse implication direction under a Noetherian target.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divided Power Algebra, proposition dpa-proposition-regular-ideal.

Source reference. Original lemma 0FJS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18456–18456. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : X \to Y$ be an immersion. If*

(1) *i is perfect,*
(2) *Y is locally Noetherian, and*
(3) *the conormal sheaf $\mathcal{C}_{X/Y}$ is finite locally free,*

then i is a regular immersion.

Proof. Translated into algebra, this is Divided Power Algebra, Proposition 0FJS. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.